Vectors and Scalars

- Dynamics – Study of motion of objects and related concepts of force and energy
- main contributors- Galileo and Isaac Newton
- Mechanics
  - kinematics
  - dynamics
- Kinematics: desription of how objects move
- Dynamics : deals with why objects move as they do

- **Mechanics :** a tool to describe the behavior of an object in physics using various elements such as displacement, velocity, and acceleration.
- **Statics mechanics** deals with when the object is not in motion,
- **dynamic mechanics** describes in detail when the object is in motion.
- **Kinematics** is the study of motion without regard for the cause. **Dynamics**: On the other hand, **dynamics** is the study of the causes of motion. This course discusses the physical laws that govern atmosphere/ocean motions.
- Kinematics + dynamics = mechanics

Physical quantities

Vectors

- Both magnitude and direction
- Eg: displacement , velocity, force, acceleration

Scalars

- Quantities that have only magnitude
- Eg: distance, speed, work, energy, power

- **Distance** is the length of the path taken by an object
- **Displacement** is the **distance** between where the object started and where it ended up(depends on direction of motion).
- Displacement is defined as the change of position of a body in a direction

For example, suppose you are travelling in a bus, 5 km east and then 3 km west. Then came back to your home through the same way. Then the total distance travelled is 8km while the total displacement is 0.

- **Speed (s)** is a scalar quantity,
  - is the rate at which an object covers distance.
  - The average **speed** is the distance (a scalar quantity) per time ratio.
  - **Speed** is ignorant of direction.
  - It is the distance travelled by a body per unit time

velocity (v) is a vector quantity; it is direction dependent.

It is defined as the displacement of a body per unit time

Unit of both speed and velocity are the same

- Acceleration (*a*)- rate of change of velocity of a body ,
- Any change in velocity(magnitude, or direction or both) brings accelerarion.
- A positive change in velocity- acceleration
- A negative change in velocity – retardation or deceleration
- If an body change its velocity from *u* to *v* then in a time *t*, the acceleration $a= (v-u)/t$
- Its dimension is L/T^2 and unit is m/s^2

- In the absence of all resistance, all objects fall with the same acceleration , in a given location. This acceleration is called acceleration due to gravity(g)

- It arises due to the force of earth's gravitation
- Its approximate value is $9.8 m/s^2$

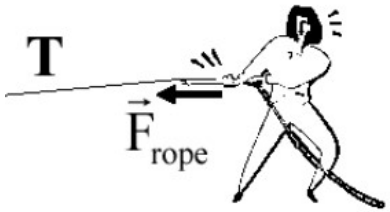
- It is an example for uniform acceleration

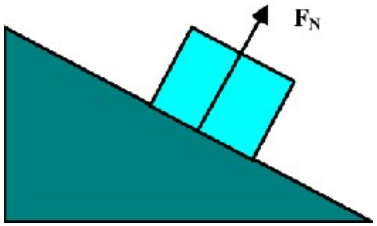
- *Isaac Newton's work represents one of the greatest contributions to science ever made by an individual.*

- Dynamics describes the relationship between the motion of objects in our everyday world and the forces acting on them
- Language of Dynamics
  - **Force:** The measure of interaction between **two objects** (pull or push). It is a vector quantity – it has a ***magnitude and direction***
  - **Mass:** The measure of how difficult it is to change object's velocity (sluggishness or inertia of the object)

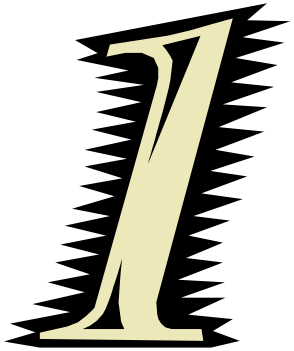
Forces

- Gravitational Force
- Archimedes Force
- Friction Force
- Tension Force
- Spring Force
- Normal Force





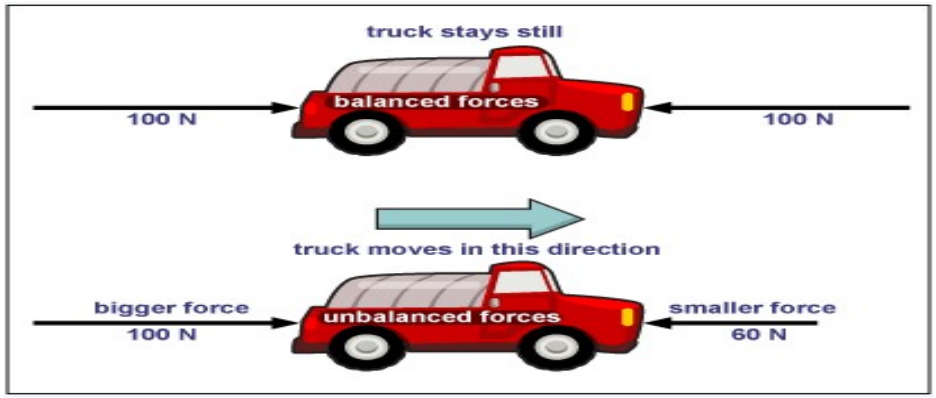
Newton's First Law
(law of inertia)



An object at rest tends to stay at rest and an object in motion tends to stay in motion unless acted upon by an unbalanced force.

Newton's first law of Motion

- An object at rest remains at rest as long as no net force acts on it. An object moving with constant velocity continues to move with the same speed and in the same direction (the same velocity) as long as no net force acts on it
- *"Keep on doing what it is doing"*
- Balanced Force



Equal forces in opposite directions produce no motion

- When forces are balanced, the acceleration of the object is zero
  - Object at rest: $v = 0$ and $a = 0$
  - Object in motion: $v \neq 0$ and $a = 0$

- The net force is defined as the vector sum of all the external forces exerted on the object. If the net force is zero, forces are balanced. When forces are balances, the object can be stationary, or move with constant velocity.

- Every object continues in its state of rest, or uniform motion in a straight line, unless it is compelled to change that state by unbalanced forces impressed upon it
- Inertia is a property of objects to resist changes is motion!
- Mass is a measure of the amount of inertia.

- Mass is a measure of the resistance of an object to changes in its velocity
- Mass is an inherent property of an object
- Scalar quantity and SI unit: kg

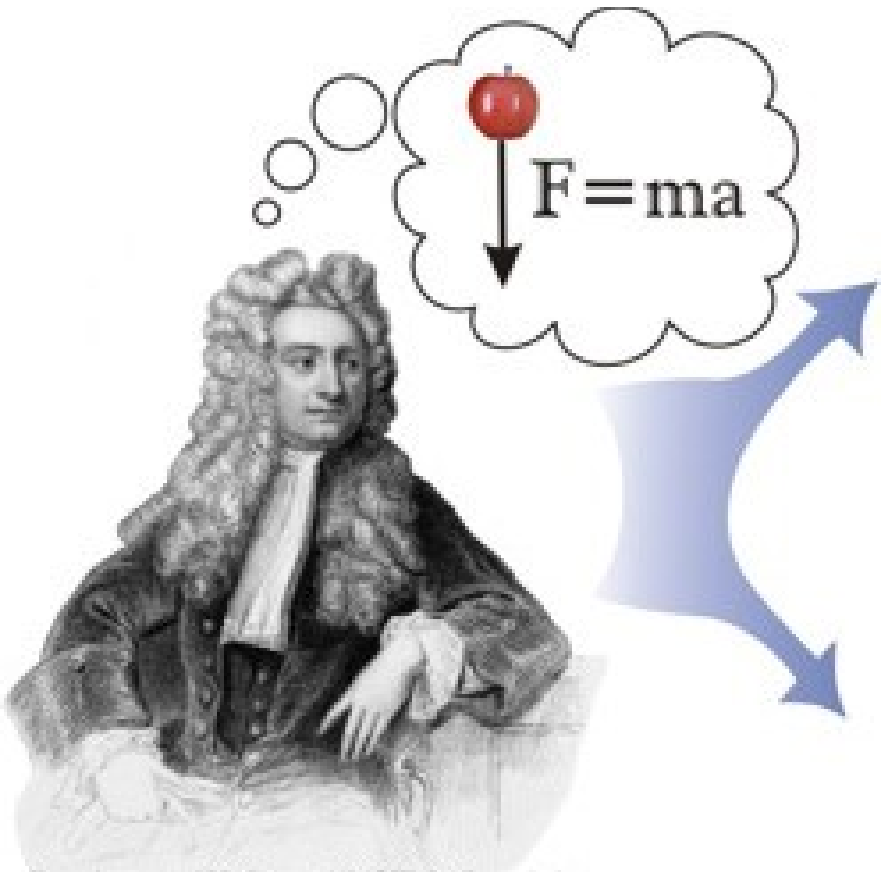
If objects in motion tend to stay in motion, why don't moving objects keep moving forever?

Things don't keep moving forever because there's almost always an unbalanced force acting upon them.

A book sliding across a table slows down and stops because of the force of *friction*.

If you throw a ball upwards it will eventually slow down and fall because of the force of *gravity*.

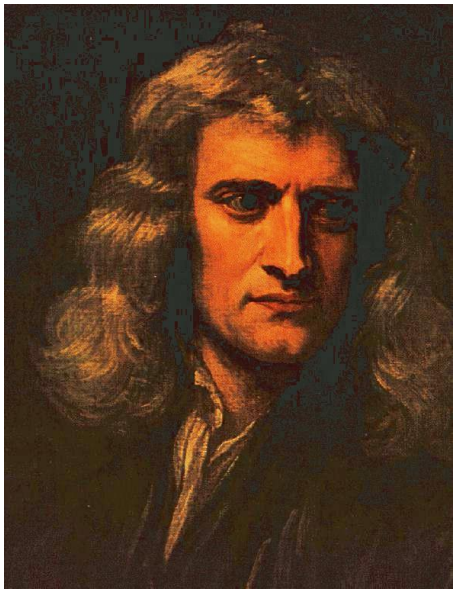
Newton's Second Law



Portrait source: U.S. Centennial of Flight Commission
http://www.centenniallofflight.gov/essay/Dictionary/newton/DI36.htm

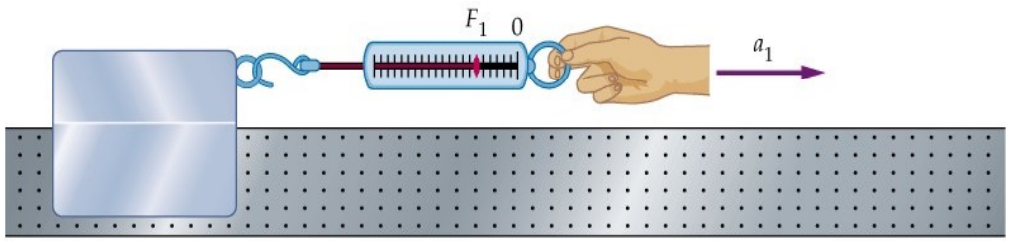
Force equals mass times acceleration.

F = ma

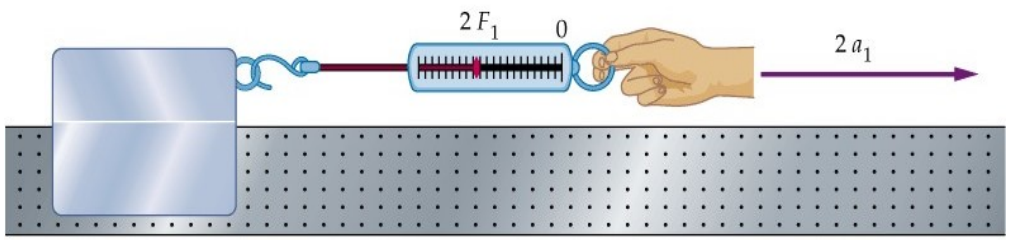


Newton's Second Law

- The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass



$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$



$$\vec{F}_{net} = \sum \vec{F} = m\vec{a}$$

Units of Force

- Newton's second law:

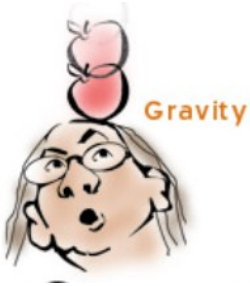
$$\vec{F}_{net} = \sum \vec{F} = m\vec{a}$$

- SI unit of force is a Newton (N)

$$1\ \mathrm{N} \equiv 1\frac{\mathrm{kg\ m}}{\mathrm{s}^2}$$

- US Customary unit of force is a pound (lb)
  - 1 N = 0.225 lb
  - Weight, also measured in lbs. is a force (mass x acceleration). What is the acceleration in that case?

Gravitational Force



- Gravitational force is a vector
- Expressed by Newton's Law of Universal Gravitation:

$$F_g = G\frac{mM}{R^2}$$

  - G – gravitational constant
  - M – mass of the Earth
  - m – mass of an object
  - R – radius of the Earth
- Direction: pointing downward

Newton's Second Law

- Force = Mass x Acceleration
- Force is measured in Newtons

ACCELERATION of GRAVITY(Earth) = 9.8 m/s^2

- **Weight (force) = mass x gravity (Earth)**

Moon's gravity is 1/6 of the Earth's

you weigh 420 Newtons on earth, what will you weigh on the Moon?

If your mass is 41.5Kg on Earth what is your mass on the Moon?

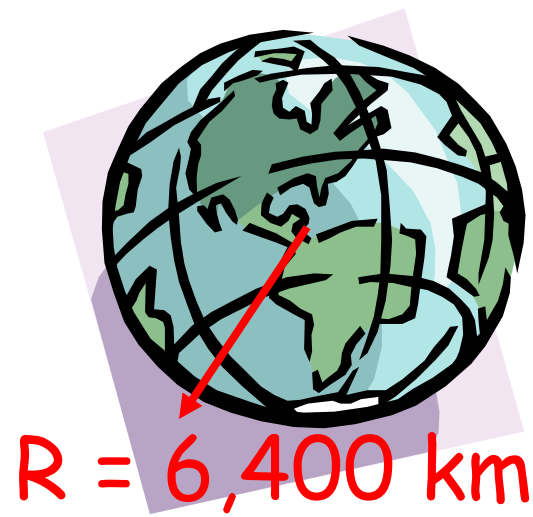
Weight

- The magnitude of the gravitational force acting on an object of mass m near the Earth's surface is called the weight w of the object: $w = mg$
- $\boldsymbol{g}$ can also be found from the Law of Universal Gravitation
- Weight has a unit of N

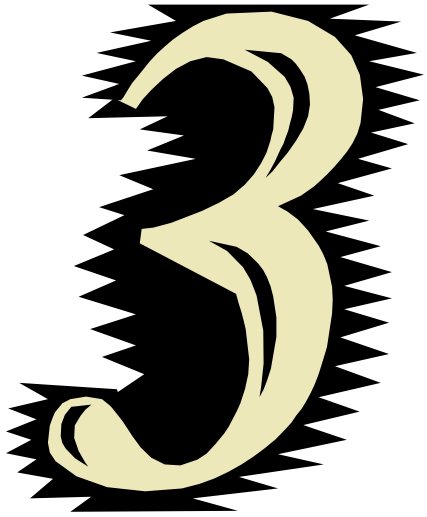
$$F_g = G\frac{mM}{R^2} \qquad w = F_g = mg$$

$$g = G\frac{M}{R^2} = 9.8\ \mathrm{m/s}^2$$

- Weight depends upon location



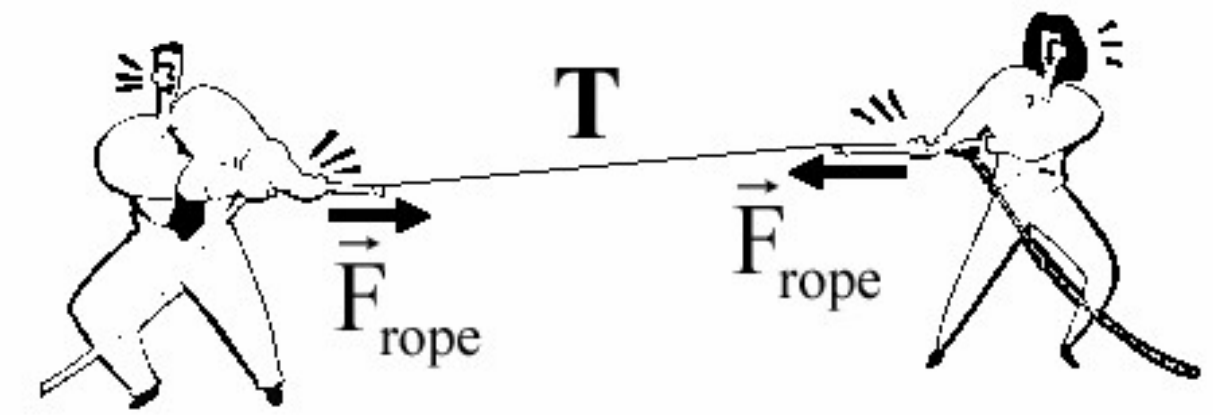
Newton's Third Law



For every action there is an equal and opposite reaction.

Newton's Third Law

- If object 1 and object 2 interact, the force exerted by object 1 on object 2 is equal in magnitude but opposite in direction to the force exerted by object 2 on object 1



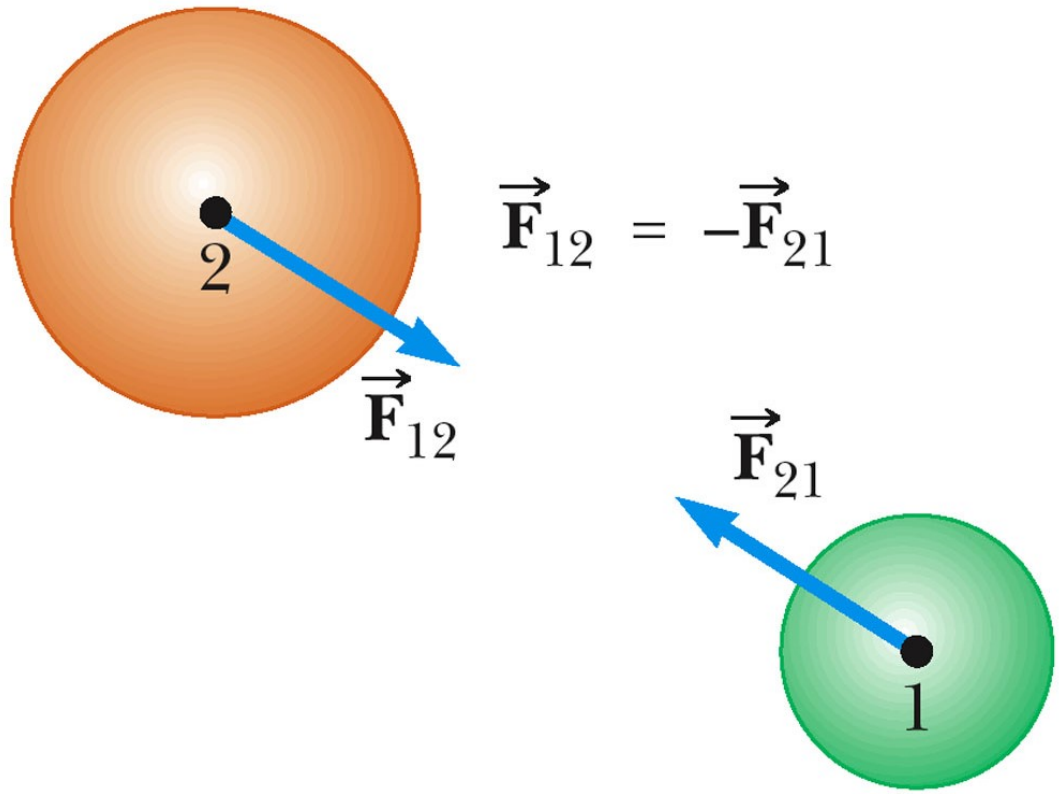
$$\left|\vec{F}_{on\ A}\right| = T = \left|\vec{F}_{on\ B}\right|$$

$$\vec{F}_{on\ A} = -\vec{F}_{on\ B}$$

- Equivalent to saying a single isolated force cannot exist

Newton's Third Law cont.

- F_{12} may be called the *action* force and F_{21} the *reaction* force
  - Actually, either force can be the action or the reaction force
- The action and reaction forces act on **different** objects



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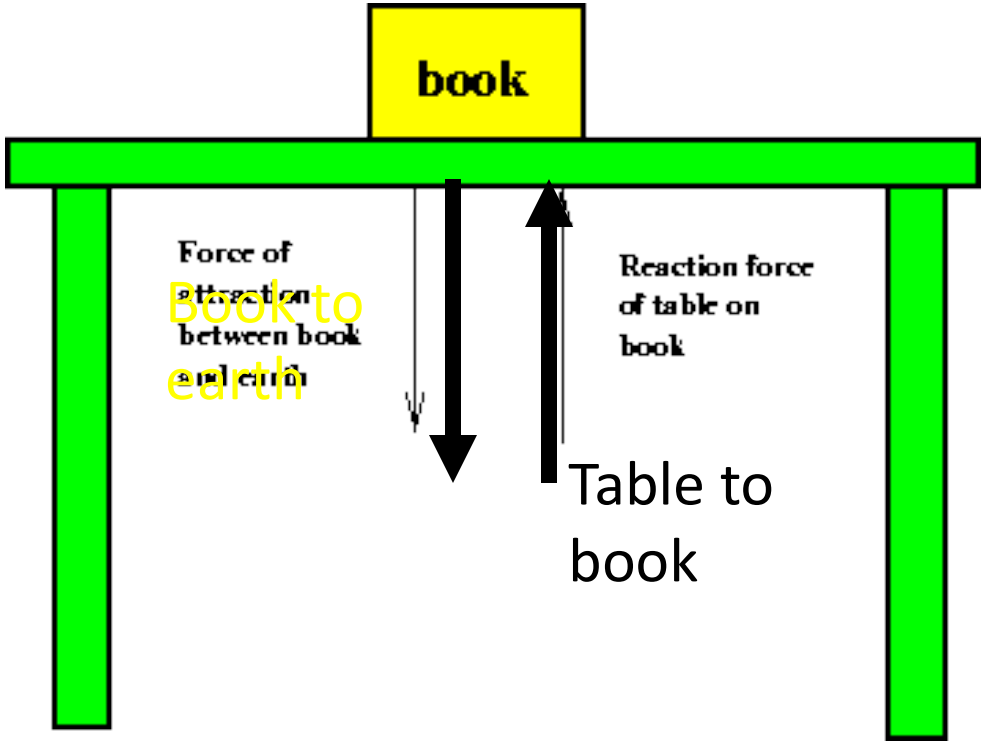
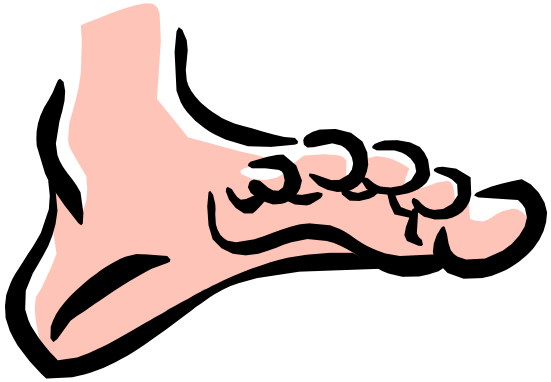
Newton's 3rd Law

- For every action there is an equal and opposite reaction.



Think about it . . .

What happens if you are standing on a skateboard or a slippery floor and push against a wall? You slide in the opposite direction (away from the wall), because you pushed on the wall but the wall pushed back on you with equal and opposite force.



Why does it hurt so much when you stub your toe? When your toe exerts a force on a rock, the rock exerts an equal force back on your toe. The harder you hit your toe against it, the more force the rock exerts back on your toe (and the more your toe hurts).

Newton’s Contributions

- Calculus
- Light is composed of rainbow colors
- Reflecting Telescope
- Laws of Motion
- Theory of Gravitation